Q1 JEE Main 2020 - 3 September (Morning)

The volume strength of $8.9 \mathrm{M} \mathrm{H_2O_2}$ solution calculated at $273 \mathrm{K}$ and 1 atm is $(\mathrm{R}=0.0821 \mathrm{L}$ atm $\mathrm{K}^{-1} \mathrm{mol}^{-1})$ (rounded off to the nearest integer)

Q2 JEE Main 2020 - 3 September (Evening)

The strengths of 5.6 volume hydrogen peroxide (of density $1 \mathrm{g/mL}$) in terms of mass percentage and molarity (M), respectively, are (Take molar mass of hydrogen peroxide as $34 \mathrm{g/mol}$)

(A) 1.7 and 0.5
(B) 0.85 and 0.5
(C) 1.7 and 0.25
(D) 0.85 and 0.25

Q3 JEE Main 2020 - 5 September (Morning)

The equation that represents the water-gas shift reaction is

(A) $\mathrm{CH_4(g)}+\mathrm{H_2O(g)} \xrightarrow[\mathrm{Ni}]{1270 \mathrm{K}} \mathrm{CO(g)}+3 \mathrm{H_2(g)}$
(B) $\mathrm{CO(g)}+\mathrm{H_2O(g)} \xrightarrow[\text{Catalyst}]{673 \mathrm{K}} \mathrm{CO_2(g)}+\mathrm{H_2(g)}$
(C) $2 \mathrm{C(s)}+\mathrm{O_2(g)}+4 \mathrm{N_2(g)} \xrightarrow{1273 \mathrm{K}} 2 \mathrm{CO(g)}+4 \mathrm{N_2(g)}$
(D) $\mathrm{C(s)}+\mathrm{H_2O(g)} \xrightarrow{1270 \mathrm{K}} \mathrm{CO(g)}+\mathrm{H_2(g)}$

Q4 JEE Main 2020 - 5 September (Evening)

Hydrogen peroxide, in the pure state, is

(A) linear and blue in color
(B) planar and blue in color
(C) linear and almost colorless
(D) non-planar and almost colorless

Q5 JEE Main 2020 - 5 September (Evening)

The one that is NOT suitable for the removal of permanent hardness of water is

(A) Calgon's method

(B) Ion-exchange method
(C) Clark's method
(D) Treatment with sodium carbonate

Q6 JEE Main 2020 - 6 September (Evening)

Dihydrogen of high purity $(> 99.95\%)$ is obtained through

(A) the electrolysis of acidified water using Pt electrodes
(B) the electrolysis of warm Ba(OH) $_2$ solution using Ni electrodes
(C) the electrolysis of brine solution
(D) the reaction of Zn with dilute HCl

Q7 JEE Main 2020 - 7 January (Morning)

In comparison to the zeolite process for the removal of parmanent hardness, the synthesis resins method is:

(A) more efficient as it can exchange both cations as well as anions4
(B) less efficient as it exchanges only anions
(C) more efficient as it can exchange only cations
(D) less efficient as the resins cannot be regenerated

Q8 JEE Main 2020 - 7 January (Evening)

Which of the following statements are correct ?
(I) On decomposition of $H_2O_2,\ O_2$ gas is released .
(II) 2-ethylanthraquinol is used in preparation of H_2O_2
(III) On heating $KClO_3,\ Pb(NO_3)_2,\ NaNO_3,\ O_2$ gas is released.
(IV) In the preparation of sodium peroxoborate, H_2O_2 is treated with sodium metaborate
(A) $I,\ II,\ IV$
(B) $II,\ III,\ IV$
(C) $I,\ II,\ III,\ IV$
(D) $I,\ II,\ III$

Q9 JEE Main 2020 - 8 January (Evening)

Hydrogen has three isotopes $(A),\ (B)$ and (C). If the number of neutron(s) in $(A),\ (B)$ and (C) respectively, are $(x),\ (y)$ and (z) the sum of $(x),\ (y)$ and (z) is :

(A) 1
(B) 2
(C) 3
(D) 4

Q10 JEE Main 2020 - 9 January (Morning)

The hardness of a water sample containing $10^{-3} M\ MgSO_4$ expressed as $CaCO_3$ equivalents (in ppm) is. . . . .
(molar mass of $MgSO_4$ is $120.37\ g/mol$)

Answer Key

Q1 (100)	**Q2** (A)	**Q3** (B)	**Q4** (D)
Q5 (C)	**Q6** (B)	**Q7** (A)	**Q8** (C)
Q9 (C)	**Q10** (100)		

Q1

Molarity of H_2O_2 solution $= 8.9M$
Volume strength of H_2O_2 solution
$= 8.9 \times 11.2 \simeq 100V$

Q2

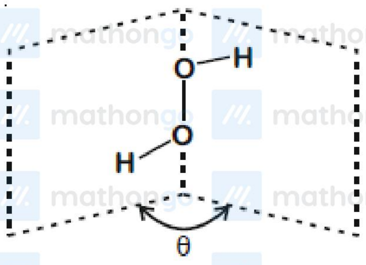
Volume strength $= 5.6$
$V.S = 11.2 \times \text{Molarity}$
Molarity $(M) = \frac{V.S}{11.2} = \frac{5.6}{11.2} = 0.5$
$0.5M \rightarrow 0.5$mole in $1L$
Mass percentage $= \frac{0.5 \times 34}{1000 \times d} \times 100$
$= \frac{17 \times 100}{1000 \times 1} = 1.7$

Q3

In water gas shift reaction, Hydrogen gas is produced economically by the reaction of carbon monoxide with water vapour at $673K$in presence of iron, cromium and copper zinc catalyst

$$CO + H_2O(g) \xrightarrow[\Delta]{\text{Catalyst}} CO_2 + H_2(g)$$

Q4



Open book like structure
- **Non planar**
- **Almost colorless**

Q5

Clark's method is used to remove temporary hardness from water

Q6

To obtain H_2 of high purity $(> 99.95\%)$ electrolysis of $Ba(OH)_2$ solution is done using Ni electrodes

Q7

Self-Explanatory

Q8

Theory Based

Q9

$$\begin{array}{llll} & {}_1^1H & {}_1^2H\,(D) & {}_1^3H\,(T) \\ \text{Number of neutrons} & & 0+1+2=3 & \\ & & 0+1+2=3 & \end{array}$$

Q10

10^{-3} molar $MgSO_4 \equiv 10^{-3}$ moles of $MgSO_4$ present in $1L$ solutions.

$n_{CaCO_3} \equiv n_{MgSO_4}$

$$ppm_{(\text{in term of } CaCO_3)} = \frac{10^{-3} \times 100}{1000} \times 10^6$$

$$ppm_{(\text{in term of } CaCO_3)} = 100\ ppm$$